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Implementing Cognitive Procurement and its Influence on Supply Chain During the Era of Digital Transformation in Oil and Gas Industry 4.0

Manoj Kannankutty, ADNOC Distribution; Aditya Manoj Menon, Abu Dhabi International School

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Abstract

The digital transformation is key for every oil and gas organization to implement its fourth industrial revolution (Industry 4.0). Digital transformation in procurement uses artificial intelligence, big data, market intelligence, etc. to create self-learning systems; for the procurement of goods and services. This is also known as cognitive procurement or procurement 4.0.

The objective of this paper is to examine the methods of implementing cognitive procurement, the impact it creates in the procurement life cycle, and in the supply chain.

The paper also provides insight into three governance models of human-machine interaction that is implemented in stages based on procurement requirement and technology adoption maturity of the organization.

- Human-in-the-loop model
- Human on-the-loop model
- Human out-of-loop model

The data for this study is obtained by preparing a list of relevant questions from multiple sources. An online survey questionnaire is sent to the procurement professionals in different organizations in the oil and gas industry. The questions reveal procurement expert's responses for:

- Potential challenges for the digital transformation in procurement and supply chain.
- Methods of enabling technologies to leverage procurement procedures and processes.

The results and the observations of the study are used to measure the benefits of implementing cognitive procurement in the oil and gas industry 4.0. The new transformed roles of procurement for industry 4.0 are also discussed in the conclusion, which includes:

- Daily activities

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- Procurement decision-making activities
- Strategic Sourcing
- Procurement to drive organizational profitability, effectiveness and efficiency

The paper will benefit oil and gas organizations that are planning to implement cognitive procurement as part of the digital transformation. The paper will also benefit any parties interested in understanding more about cognitive procurement.

Keywords: Cognitive Procurement, Artificial Intelligence, Digital Transformation, Oil and Gas Industry 4.0, Strategic Sourcing

Introduction

Oil and Gas Industry Outlook. The oil and gas industry is going through a period of uncertainty and volatility in the demand contributed my multiple factors including the impact of COVID 19, development in alternate energy options and global geopolitical implications. The below Fig.1 and Fig. 2 show the global demand forecast for liquid fuels consumption and the estimated demand and supply balance.

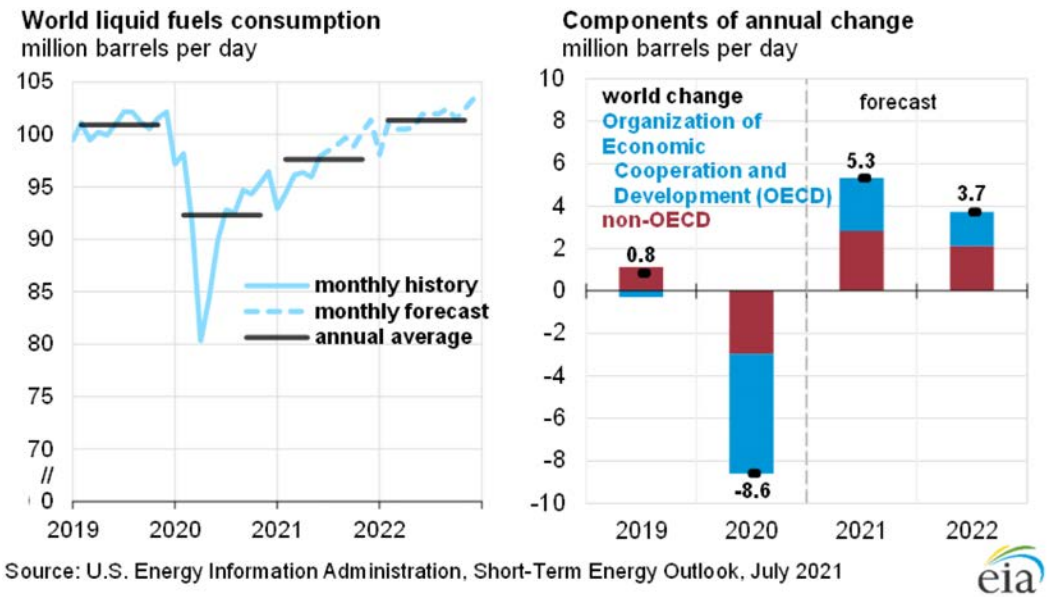

Figure 1—Energy Information Administration (EIA) short-term global demand forecast

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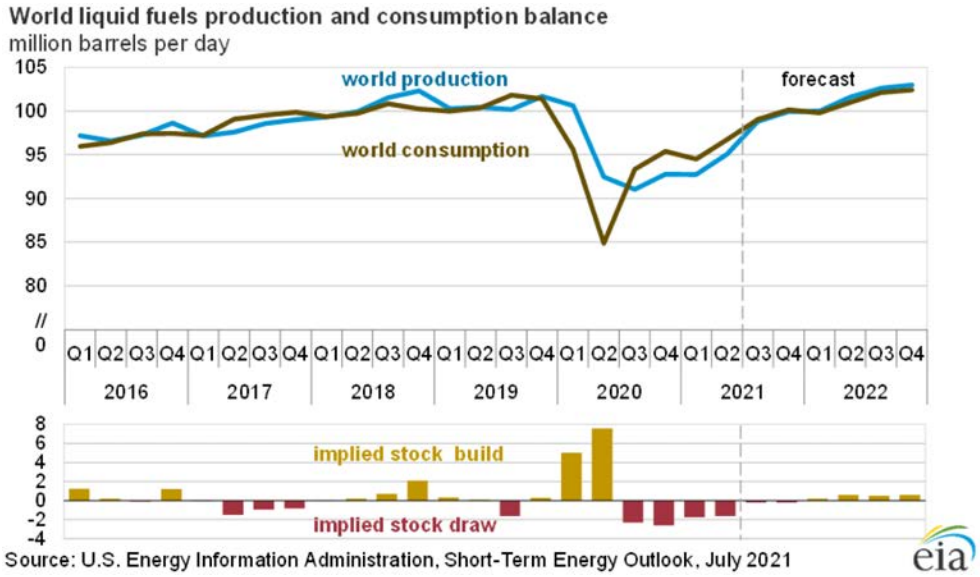

Figure 2—EIA short time forecast of fluctuations in the demand and supply balance

It is very important for oil and gas companies to find ways to improve operational efficiency and reduce expenses. Optimizing oil and gas procurement strategies and best industry practices can contribute to significant improvements in operational and financial efficiency.

To do this, we need to examine current procurement processes in oil and gas companies and understand how digital technologies are transforming procurement and how these digital technologies are being used to improve the bottom line of oil and gas procurement.

Oil and gas companies can develop a new digital transformation strategy or include procurement as part of other digital transformation initiatives for production, sales and marketing.

Oil and Gas Industry 4.0

Oil and Gas 4.0 is where the oil and gas industry meets the fourth industrial revolution. This is an era where digital innovation and technology transformation are used to drive economic progress and demand for every product in the oil and gas industry.

The fourth industrial revolution is driven by the convergence of various digital technologies to create optimized and efficient production, sales, marketing, distribution, and customer experience. This is having a major impact on customer behavior.

Digital Technologies that accelerate Oil and Gas Industry 4.0

There are many technologies that are accelerating fourth industrial revolution in the oil and gas industry. These technologies are also providing the infrastructure for the implementation of procurement 4.0. Some cutting edge digital technologies that will accelerate the oil and gas industry 4.0 are discussed below.

Artificial Intelligence (AI)

AI is the simulation of human intelligence processes by the machines. The basic objective of AI is to enable computers to perform intellectual tasks such as decision-making and problem solving with intelligence faster and better than human beings.

There are four types of AI based on the properties (Dontha 2020) as shown in in Fig. 3

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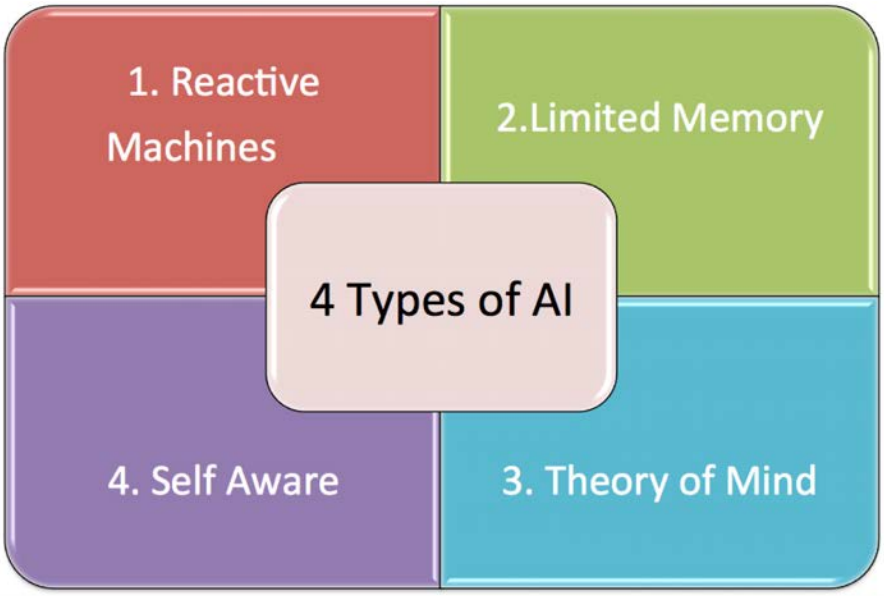

Figure 3—Four types of AI

Reactive Machines (RM). Reactive Machines respond to an input with specific output. The model has no memory and does not perform any learning.

Limited Memory (LM). Limited memory types refer to an AI that is able to learn from historical data and make better predictions. Most current applications fall into this category. The use of this AI in procurement 4.0 will help in making unbiased decision in procurement.

Theory of Mind (TM). This type of AI exists in theory or development is underway. A Theory of Mind AI will interact with the emotions and thoughts of the humans. This type of AI will be able to understand human emotions and respond accordingly, just like any other emotional human.

Self Aware (SA). The SA is the final stage of AI that has evolved into a human-like being with independent intelligence. This type of AI is still in development and the impact of this type on the human race needs to be analyzed before looking into it.

Three types of AI based on Maturity Stages (Advani 2021). AI can be classified into three different levels of solutions according to their maturity. Fig. 4 shows three types of AI belonging to the 3 maturity levels.

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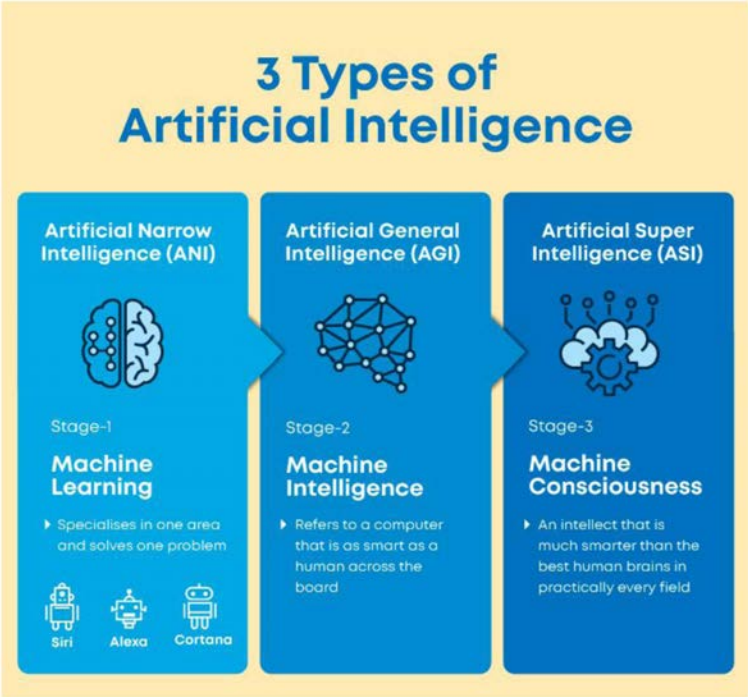

Figure 4—Three types of AI based on Maturity Stages

Artificial Narrow Intelligence (ANI). ANI is designed to solve a single problem. Since it has limited capabilities, it can be executed well for a particular situation. ANI provides the e-commerce customer with a product recommendation based on the history of search and purchase behavior. This is the only type of Artificial Intelligence that is mature today. ANI is able to approach or even surpass the human way of working in very specific contexts with greater speed.

Artificial General Intelligence (AGI). AGI is defined as AI that has human-level cognitive functions. An AGI system can consist of hundreds of ANI systems to mimic human brain function. A study on the world's best AI solution platform reveals it takes about 40 minutes to simulate the activity of a single neuron (Advani 2021).

Artificial Super Intelligence (ASI). An ASI system is superior to AGI which would be able to surpass all human capabilities. This would include better and faster decision making than human brain.

Machine Learning (ML) (Advani 2021)

Machine learning is a subset of AI that analyzes historical data to identify patterns thus reaching conclusions without relying on human expertise. With automated analytics, businesses can save time and make better decisions

Deep Learning (DL) (Advani 2021)

Deep Learning is an ML technique. It teaches a machine to process inputs through layers in order to classify, infer and predict the outcome.

Neural Networks (NN) (Advani 2021)

NN work on the similar principles as of human neural cells. They produce a series of algorithms that simulate the behavior of a human brain by capturing the relationship between variables in various contexts.

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Natural Language Processing (NLP) (Advani 2021)

Natural Language Processing employs Computational Linguistics (CL) together with machine learning, statistical, and deep learning models to read, understand, and interpret natural language.

Chat bots use natural language processing to interpret the intent of the user and respond accordingly. They rely on NLP to accomplish their tasks. NLP in conjunction with other AI subsets, can save significant time for contract management by reviewing legal contracts and verifying the compliance.

Computer Vision (CV)

Computer vision uses algorithms to interpret various attributes of an image by breaking down the image and studying different parts of the objects. This helps the machine to analyze, classify and learn from a set of images, to make a better output decision based on previous observations. A good example for computer vision is Google's vision API that is used to understand and interpret the attributes of image.

Computer vision could help approve payment as per the actual progress for a construction project based on the real time progress image obtained from the construction site.

Big Data

Big data is a combination of structured, semi structured and unstructured high-volume data collected form internal and external sources.

Big data is often characterized by the three V's (Emmanuel et al. 2016):

- The large **Volume** of data from multiple sources
- The wide **Variety** of data types frequently stored in big data systems
- The **Velocity**, at which much of the data is generated, collected and processed.

In a nutshell, big data is a solution for analyzing huge volume of structured, semi-structured and unstructured data from various sources with variety of data types which are generated, collected and processed at high velocity.

The big data have major role to play in successful implementation of procurement 4.0.

Internet of Things (IoT)

The Internet of things (IoT) (Xia, F et al. 2012)is a system of interrelated computing devices, mechanical and digital machines, objects, animals or human beings that are embedded with sensors, software, and other technologies to obtain unique identifiers for the purpose of connecting and exchanging data with other devices and systems over the Internet.

Robotic Process Automation (RPA)

Robotic process automation (van der Aalst et al. 2018) is a software technology that helps to build, deploy, and manage software robots that emulate human actions. As an example, RPA can be used to read paper-based supplier delivery notes and enter into the system without human interaction.

Blockchain

Blockchain uses decentralized ledger technology for recording transactions by using a chain of blocks. Each block can hold a transaction and copies of the transaction are available to all participants of the blockchain. The new blocks are added to the chain by using hash created from the last block. This way the transactions are irreversible and secured in the block.

Blockchain can be used in procurement (King 2020) to record secure transactions for tracking a product or transaction from source of acquisition to payment. A new block is added to the chain for each transaction

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down the supply chain. Many innovations can be initiated in procurement processes and supply chain management using blockchain technology.

Cloud Computing

In Cloud Computing (Armbrust, M. et al. 2010), software, analytics, cyber security, and intelligence are delivered over the Internet to allow quicker innovation, flexible resources, the exchange of data, and cost savings.

This eliminates the need for companies to invest in IT infrastructure data centers and networks. In Cloud Computing, the subscriber pays only for cloud services they use, helping to lower their operating costs, run their infrastructure more efficiently and scale as their business needs change.

Cognitive Procurement

Digital transformation in procurement uses artificial intelligence, Big Data, market intelligence, etc. to create self-learning systems; for procurement of goods and services. This is also known as cognitive procurement or procurement 4.0.

Digital transformation in procurement is as important to oil and gas companies as transformation efforts in manufacturing, marketing and sales. This is simply because any savings in procurement has a huge impact on the oil and gas company's financial bottom line.

Currently, the procurement department of most oil and gas companies relies on the expertise of procurement and business staff to identify suppliers for procurement needs. The organization manages purchasing and contracts mainly using internal data from the organization's Enterprise Resource Planning (ERP) system.

However, most oil and gas companies do not have mandatory policies to collect Market Intelligence (MI) and Big Data from external sources to obtain insight of alternative suppliers, alternative supply chain, cost, value, quality of service delivery, etc. before issuing purchase orders or awarding contracts. Such information benchmarked against competitors at regional and global levels provides the company with additional insights for better decision-making before awarding contracts or purchase orders.

The oil and gas companies can use cognitive computing to provide faster and more accurate procurement analytics. This will also lead to improved operational efficiency and tactical purchasing. Cognitive computing should also help to identify supplier risks and opportunities. Fig. 5 illustrates cognitive computing in procurement.

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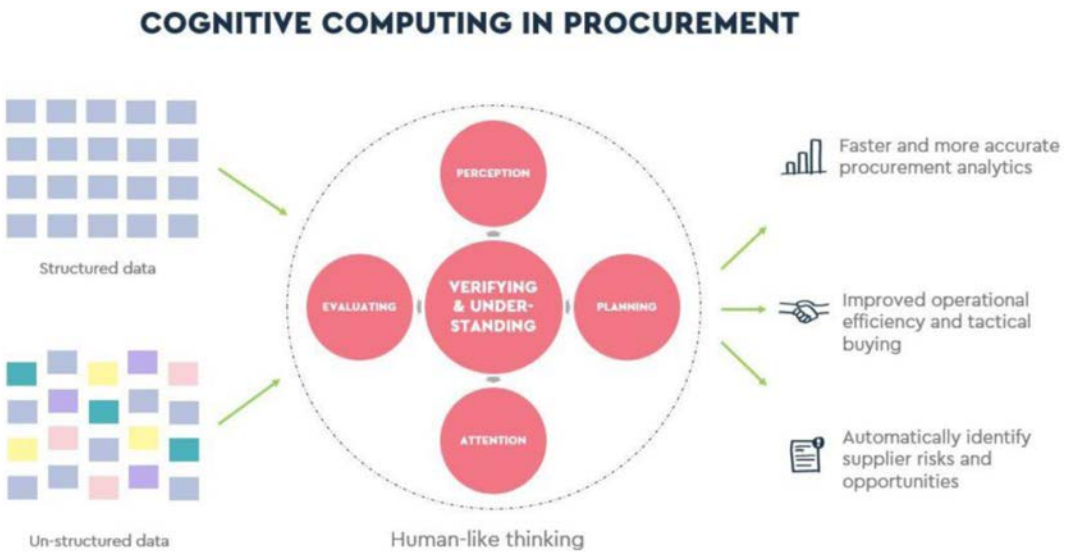

Figure 5—Cognitive computing in procurement. The above image is taken form https://sievo.com/resources/ai-in-procurement. Accessed on 15 July 2021.

Companies that successfully implement Procurement 4.0 experience competitive advantages by building closer, more strategic relationships with suppliers.

The Fig. 6 shows how modern digital technologies transform various stages of procurement lifecycle.

Figure 6—AI in Procurement Software. The above image is taken form https://sievo.com/resources/ai-in-procurement. Accessed on 15 July 2021.

Supplier Risk Management

Artificial Intelligence, Big Data and Blockchain technologies can be used to monitor and identify potential risk exposures throughout the supply chain. For example, during the COVID-19 pandemic, many oil companies experienced delays in the delivery of critical products because their current supply chain depends on many components manufactured in China. Some companies that did not source components directly from China also experienced delays in sourcing products because their subcontractors sourced those components

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from China. To make production fail-safe, oil and gas companies should have access to many data sources to have information on alternative products, suppliers, manufacturing countries, market information, probable supply chain delays that affect the procurement of critical components. This way, the organization can be prepared with risk mitigation plans to handle the situations without any major loss of production.

AI in procurement
Artificial intelligence can be used to automatically review and approve purchase orders.

Automation of payments
Machine learning can be used for automated payment approvals to reduce the invoice payment cycle.

AI Spend Analysis Software
In purchasing, ML algorithms can be used in software for faster processes such as classifying spend and matching with suppliers.

New supplier identification
Big Data techniques are enabling new ways to identify, manage and use supplier data in public and private databases. Market intelligence from multiple sources combined with Big Data and machine learning algorithms can improve the identification of new suppliers for procurement needs.

AI in Strategic Sourcing
Artificial Intelligence along with Big Data, market intelligence can act as a sourcing advisor. This can help procurement to get better insight for better negotiations and informed decisions.

AI in contract management
Natural language processing combined with machine learning can check contract documents for legal compliance. The AL can report potential compliance violations and the expert can verify and confirm them. This significantly shortens contract review time and reduces contract management costs.

Methods for implementing cognitive sourcing

Human On The Loop (HOTL)
Human on the Loop is a primitive process that most companies still use. This is due to the static nature of procurement in the organizations. In this process, the procurement specialist only makes predictions and decisions about which external source is best for procuring goods, services or labor. Therefore, there is no AI system that reviews the data and helps the procurement specialist make an informed decision. Instead, specialists must review the data themselves to shortlist procurement providers before making a final, informed decision. The disadvantage of this type of system is that humans are prone to errors and inaccuracies when it comes to analyzing the data. There is also the possibility of vested interests, which could lead to a biased decision that is not necessarily beneficial to the company.

Human In The Loop (HITL)
HITL in procurement is a process in which internal and external procurement related data is fed into an AI system. The system, makes prediction on the most suitable supplier for acquiring goods, services, or work. However, in this case the AI system suggests sourcing decision insights procurement specialist of the firm. The AI system only acts as an aiding system, which gives a short list of external sources, that are most cost-efficient, based on the data analyzed by the system. The final decision is made by the procurement specialist and not by the AI system. This can be advantageous because the data collected can have some degree of uncertainty, that would make the predictions not 100% accurate. Therefore, the specialists can review the

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predictions made by the AI system to make a more informed decision. This would help improve the accuracy of decision making since AI systems analyze the data in an unbiased manner and the specialists can come to the final decision by reviewing the short-list of external providers the AI system has come up with.

Studies conducted by Stanford in 2018 found that HITL is more accurate and cost-efficient than either only AI system or only procurement specialist (Wolfewicz 2021).

Human Out Of The Loop (HOOTL):
HOOTL in procurement is the process by which the AI system makes prediction and final decision as to which external source is most suitable for acquiring goods, services, or work. Therefore, there no involvement of procurement specialists. This is most advanced model and can be implemented when AI stage matures form ANI to AGI.

Key Barriers on Procurement 4.0 implementation in Oil and Gas Industry and how to overcome them

Even though the benefits of procurement 4.0 are clear for oil and gas companies, these do not guarantee that procurement teams have an easy task implementing them. Even with management buy-in and enough allocated budget, the organization will find it challenging to transform and implement procurement 4.0.

It is important to identify the key barriers in procurement 4.0 and how to overcome them. After a careful study and analysis within the oil and gas group of companies, following are required to overcome the potential barriers for the successful implementation of procurement 4.0 as part of implementing industry 4.0 digital transformation program.

Digital Transformation Strategy. The basic requirement for oil and gas companies is a clear digital transformation strategy for implementing procurement 4.0. This needs strong backing and support from the board. The chief executive management should take the ownership of monitoring and executing the digital transformation strategy. A survey from McKinsey & Company, (Fig. 7) states successful digital and analytics transformation are focused and well defined.

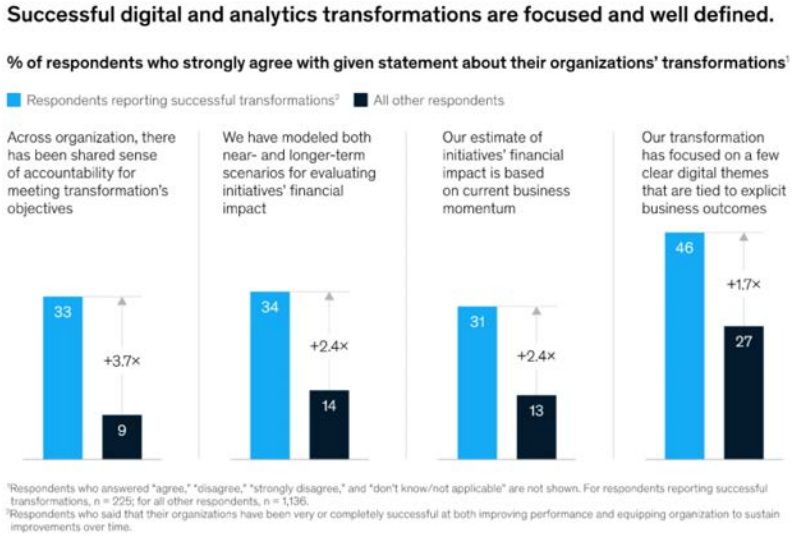

Figure 7—Digital Transformation statistics. The above image is taken from https://www.mckinsey.com/business-functions/organization/our-insights/unlocking-success-in-digital-transformations#. Accessed on 24 July 2021.

Digital Transformation Program. It is advisable to have a clear execution program led by the Transformation Office. The Transformation Office should oversee all project initiatives of the digital

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transformation program. Typically, the Digital Transformation Office (DTO) manages the initiatives and projects required for Procurement 4.0 transformation and is implemented in close collaboration with the Chief Executive Officer. Fig. 8 shows the results of a survey conducted by McKinsey & Company on the roles and responsibilities of strategic and executive leadership for the successful execution of the digital transformation program.

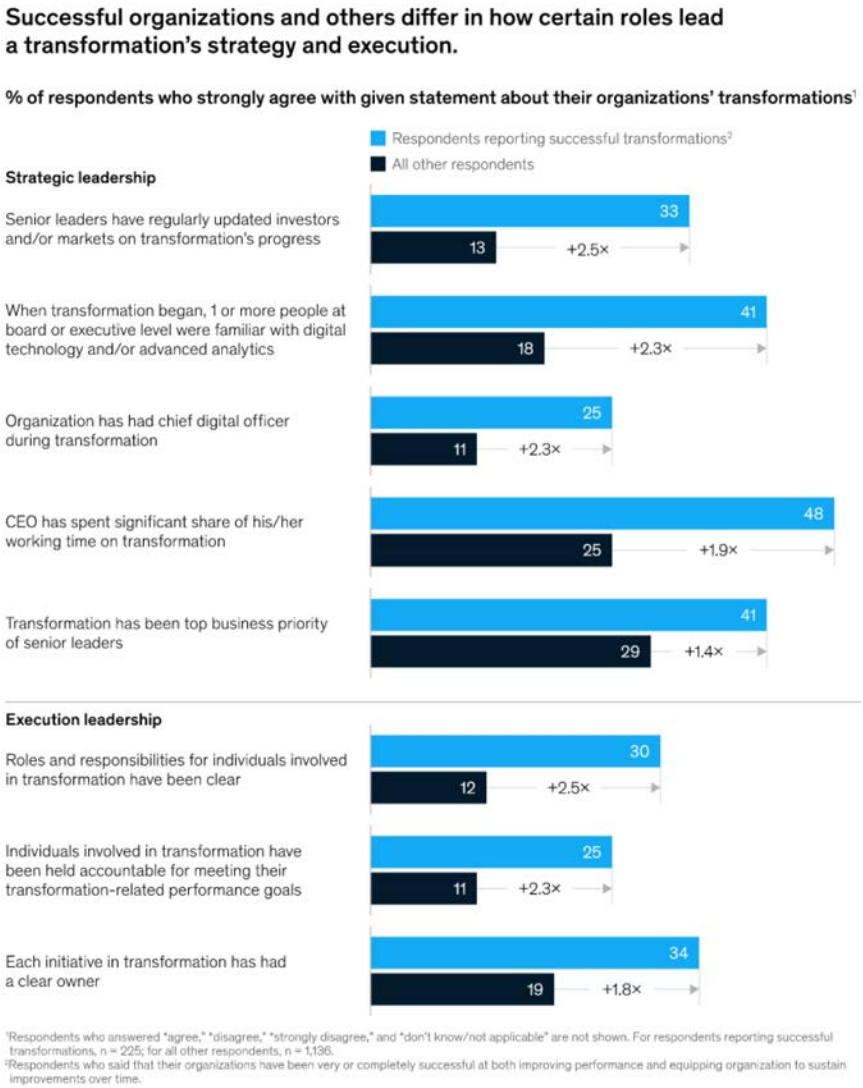

Figure 8—Survey on the roles of strategic and execution leadership on successful transformation strategy and execution. The above image is taken from https://www.mckinsey.com/business-functions/organization/our-insights/unlocking-success-in-digital-transformations#. Accessed on 24 July 2021.

Organizational Culture. Organizational culture and mindset need to be aligned with the new ways of working as a result of digital transformation implementation. A successful organization will have more empowered employees and will foster adoption of agile methodologies in the creation and execution of the transformation projects. Fig. 9 shows the results of a survey conducted by McKinsey & Company, in which the majority of respondents believe that an enterprise-wide scope is much more likely to lead to a successful transformation than companies in which only a specific business unit is involved in the transformation.

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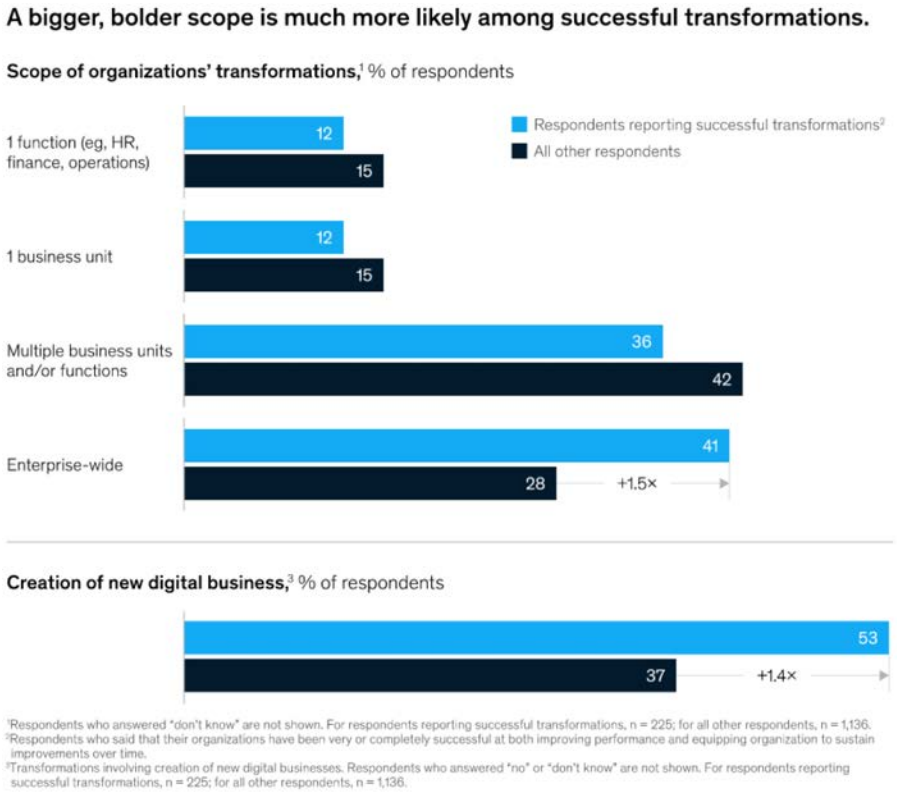

Figure 9—Survey on probable scope for successful transformation. The above image is taken from https://www.mckinsey.com/business-functions/organization/our-insights/unlocking-success-in-digital-transformations#. Accessed on 24 July 2021.

Digital Transformation Operational Model. Before initiating projects for the digital transformation program, it is always important to develop digital transformational operating model. The projects and programs to drive digital transformation are derived from the digital transformation operational model.

Change Management. To drive digital transformation program, it may be required to overcome different organizational barriers. For this, the organization may have to make changes to different organizational assets as a part of digital transformation. Following are some of the key barriers which may need to be addressed for successfully implementing procurement 4.0.

- Readiness of infrastructure for Procurement 4.0
- Restructuring of human resource
- Empowerment of innovation
- Suppliers as partners for transformation
- Lean communication hierarchy
- Digitization risk management
- Organizational digital capability
- Adoption of agile methodology

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Description of Methodology

A questionnaire was sent out to collect primary data from employees working in procurement or related business departments in various oil and gas companies. The questionnaire has used a five-point Likert agreement scale. The survey questions cover the following areas (Bienhaus, F. et al, 2018):

1. Impact of digital transformation
2. Potential challenges in digital transformation of procurement and supply chain
3. Enabling technology to leverage procurement

The findings and observations of the study are used to measure the benefits of implementing cognitive procurement in the oil and gas industry 4.0. The new transformed roles of procurement for industry 4.0 are also discussed in the conclusion, which includes:

- Daily activities
- Procurement decision-making activities
- Strategic Sourcing
- Procurement to drive organizational profitability, effectiveness and efficiency

Survey Data and Results

Data for this study were obtained by compiling a list of relevant questions from multiple sources. An online survey questionnaire was sent to procurement professionals in various organizations in the oil and gas industry. The results were presented based on the percentage of respective responses from the participants for each questionnaire conducted as part of the study.

The survey on the impact of digital transformation (Fig. 10) on procurement was conducted with 5 questions. 88.9% of the participants either agreed or strongly agreed that procurement will transform into a strategic and innovative function. Procurement will be a function to support efficiency, effectiveness and profitability. 92.6% of participants either agreed or strongly agreed that Integrating Market Intelligence data from various external sources for procurement will give the organization better insight for procurement decisions. Majority of the participants also agree or strongly agree that artificial intelligence will be able to reduce day-to-day procurement activities and 70.9% of the participants believe that AI can take over the procurement process in the long run. In summary, the survey result shows that digital transformation will impact the procurement function of oil and gas companies.

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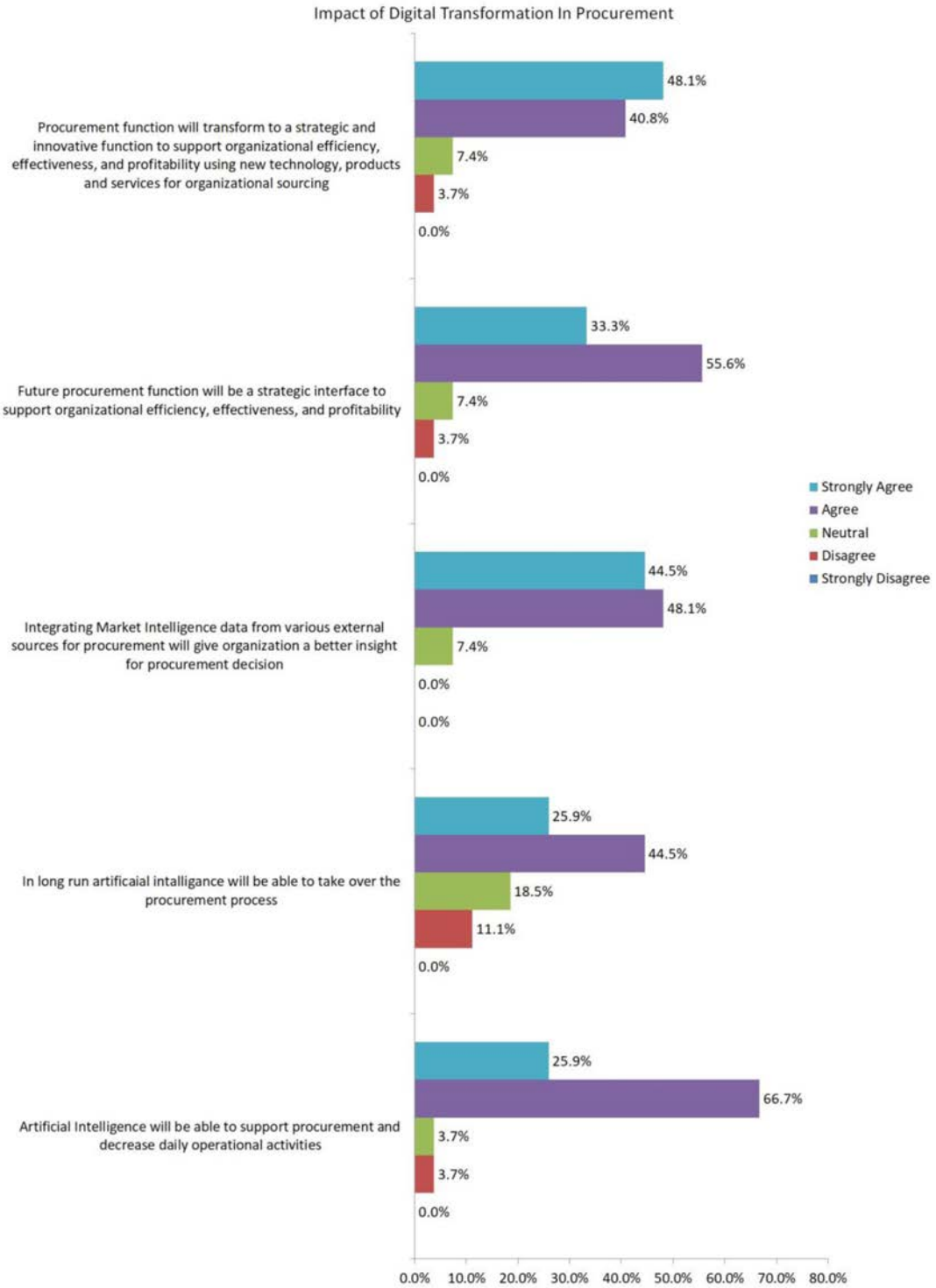

Figure 10—Survey on Impact of digital transformation on procurement

In this figure below (Fig. 11), there are major differences in the results compared to the first survey figure (Fig. 10). The survey points to skills gaps, inadequate infrastructure, changing job roles, and organizational communication structure as potential barriers to implementing digital transformation. For example, a common problem we identify is that the existing infrastructure in the organization cannot handle digital transformation. Therefore, this existing infrastructure should be upgraded so that it can support digital transformation. The skill gaps of the employees need to be addressed through training and hiring new employees. Also, the organization needs to be restructured to support digital transformation.

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However, the majority of participants either agree or strongly agree that the organization's leadership supports transformation, creativity, and innovation. They also agree or strongly agree that suppliers are aligned with the digital transformation process. The majority of respondents also agree or strongly agree that their organization has a clear digital strategy and has established a transformation office with the right risk management tools.

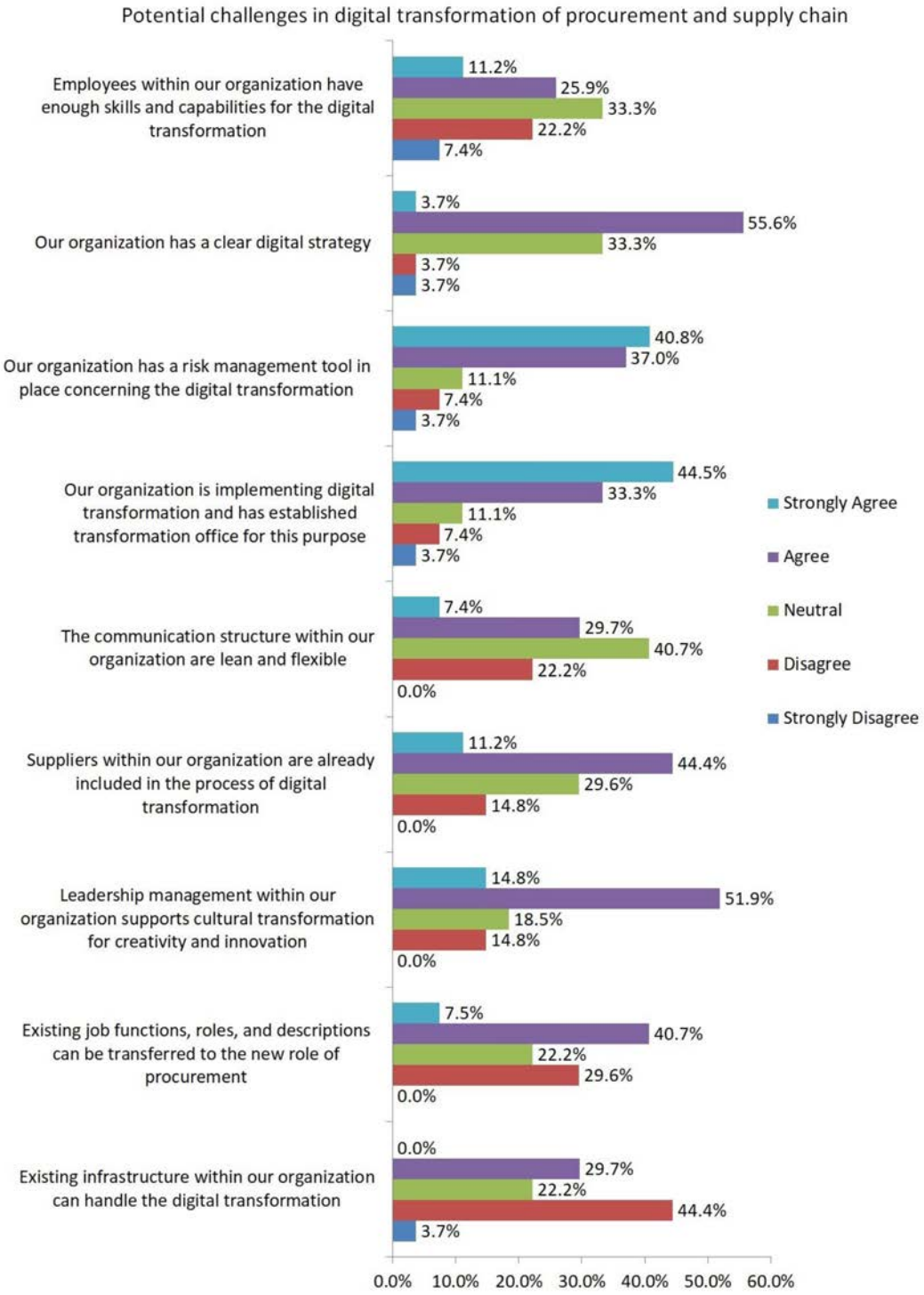

Figure 11—Survey on Impact of digital transformation on procurement

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The Fig.12 reveals whooping majority of the participants agree leveraging AI, ML, Block Chain, Big data analytics, predictive analytics tools are going to speedup, simplify and increase the transparency of procurement and supply chain ecosystem. They also agree cyber security implementation in supply chain requires a cross-functional approach. It is important to highlight that none of the participants disagree or strongly disagree to any of the 6-questionnaire related to enabling digital transformation technologies to leverage the procurement.

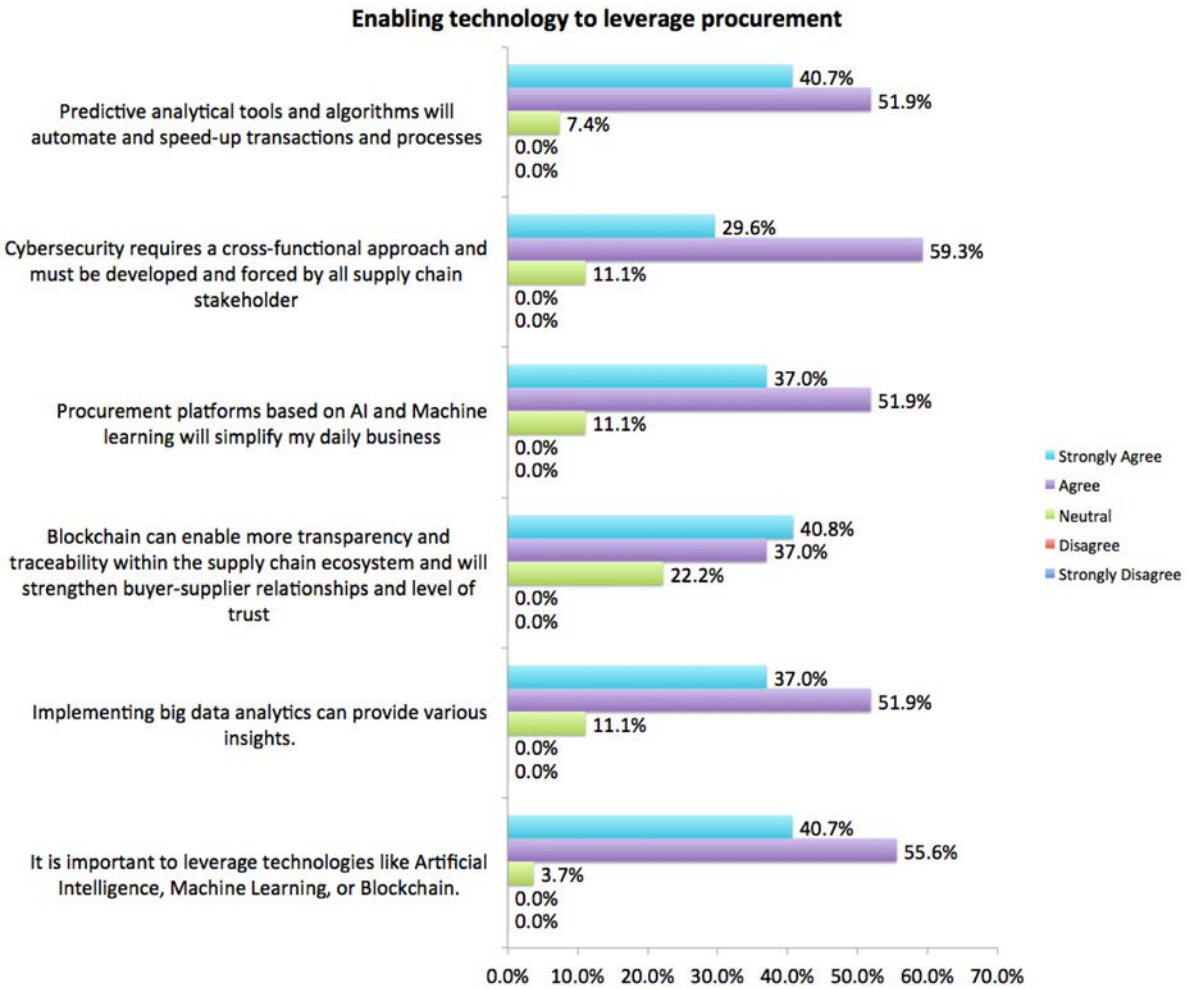

Figure 12—Survey on Impact of digital transformation on procurement

Conclusion

It has to be agreed with the majority view the daily activities of procurement is simplified and automated by the use of digital transformation technologies. The big data, with connectors to loads of unstructured and structured external sourcing data, internal sourcing data and predictive analytics will simplify the procurement decision-making process. The block chain will improve the transparency of the supply chain.

The digital transformation will transform procurement function of oil and gas industry to a strategic, profitable, efficient and effective function within the organization. This shall support procurement from a business support function to a function capable of creating positive influence in financial bottom line for the companies.

The innovation using convergence big data, AI, ML, block chain can create cognitive procurement models which could be used initially as human-in-loop model. As the technologies further matures, there

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is possibility of some oil and gas organizations to use human-out-of-loop for the procurement processes. However, time will have to decide such a scenario for procurement.

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List of Abbreviations

AGI	Artificial General Intelligence
AI	Artificial Intelligence
ANI	Artificial Narrow Intelligence
ASI	Artificial Super Intelligence
CL	Computational Linguistics
CV	Computer Vision
DL	Deep Learning
DTL	Digital Transformation Office
ERP	Enterprise Resource Planning
HITL	Human In The Loop
HOTL	Human On The Loop
HOOTL	Human Out Of The Loop
IoT	Internet Of Things
LM	Limited Memory
ML	Machine Learning
MI	Market Intelligence
NLP	Natural Language Processing
NN	Neural Networks
RM	Reactive Machines
RPA	Robotic Process Automation
SA	Self Aware
TM	Theory of Mind

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